

Randomized and low-rank approximation

Sharp gap between volume sampling and the worst matrix with a prescribed spectrum

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging because the adversarial eigenvector choice and optimal subset must be compared sharply; community impact is spectrum-sensitive Nyström and column selection.

Further partial resolution — all positive spectra, 14 September 2026

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. [Verified affiliation and submission record.](#)

[Theorem 1.1](#) proves, for every positive spectrum and $2 \leq k \leq n - 2$,

$$R_{n,k} \leq \min\{k + 1, n - k, 2^{34} e^{2560} \sqrt{k}\}.$$

The limiting-profile estimate and exact-spectrum reduction retain the real orthogonal model and the original maximum-before-minimum quantifiers. The enormous universal constant is an asymptotic bound, not a practical estimate. **The sharp joint dependence on n, k remains open:** no matching growing lower bound or all-spectrum square-root-complement-size bound in the strongly unbalanced regime is established.

A separate [independent Codex AI-agent audit](#) passed this partial theorem, and 16 diagnostic groups passed in a documented scratch rerun. **Status remains Partially resolved.** The existing one-column result and original target below are retained. [Proof source.](#) AI assistance is disclosed; no Lean verification, external human peer review, formal verification or novelty claim is asserted.

Partial resolution: the one-column case, 12 September 2026

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[Theorem 1 and Sections 2-3 of the note](#) prove $x_1(\lambda) = y_1(\lambda)$ and therefore $R_{n,1} = 1$ for every $n \geq 3$ and every positive spectrum. The proof preserves the original real orthogonal model and max-min quantifier order: a zero-diagonal orthogonal basis for a trace-zero symmetric matrix equalizes all one-column residuals.

A separate [independent Codex AI-agent review](#) returned **PASS for this special case**. The joint dependence for $2 \leq k \leq n - 2$, $n \geq 4$, remains unresolved; this is not a full solution. The elementary identity is recorded without a novelty or priority claim. The original target and ID are unchanged, and the historical checks below are retained. AI assistance was used for review and submission preparation; informal automated review is not external human peer review or formal verification. No Lean verification was performed. [Proof source](#) · [Submission record](#).

Problem statement

For $n \geq 3$, $1 \leq k \leq n - 2$, and $\lambda \in (0, \infty)^n$, write $K_V = V^T \text{diag}(\lambda) V$, $V \in O(n)$, and define

$$x_k(\lambda) = \max_{V \in O(n)} \min_{|I|=k} \text{tr}(K_V - (K_V)_{:,I} (K_V)_{II}^{-1} (K_V)_{I,:}), \quad y_k(\lambda) = (k + 1) \frac{e_{k+1}(\lambda)}{e_k(\lambda)},$$

where $e_j(\lambda) = \sum_{|I|=j} \prod_{i \in I} \lambda_i$. Determine, up to universal multiplicative constants, the dependence on n, k of

$$R_{n,k} = \sup_{\lambda \in (0,\infty)^n} \frac{y_k(\lambda)}{x_k(\lambda)}.$$

Here y_k is the expected trace error when the selected subset has probability $\det((K_V)_{II})/e_k(\lambda)$. The question compares that expectation with optimal subset selection after an adversary chooses the eigenvectors. This order of quantifiers is essential. The endpoint $R_{n,n-1} = 1$ is known and excluded. The displayed supremum is a concrete subquestion of the source's request for spectrum-dependent tightness bounds; stronger bounds retaining the full spectrum would also be valuable.

References

Amsel et al., [workshop report](#), §4.2, Problem 4.7 and equations (15)–(16). Mark Fornace and Michael Lindsey, *Column and Row Subset Selection Using Nuclear Scores: Algorithms and Theory for Nyström Approximation, CUR Decomposition, and Graph Laplacian Reduction*, §4 and Appendix C, for determinantal sampling and the elementary-symmetric-polynomial formulas.

Status check

Searches included "minimax" "volume sampling" "2026", "volume sampling" "worst" "spectrum" "2026", and "volume sampling" "tightness" Fornace. No matching sharp estimate for $R_{n,k}$ was located. Generic column-subset approximation guarantees do not by themselves settle this spectral minimax ratio.

Audit — 2026-09-10

Rechecked [workshop Problem 4.7 and equations \(15\)–\(16\)](#). They support the stated spectral-minimax subquestion. Volume-sampling, worst-spectrum, and tightness searches found no sharp joint estimate for the displayed ratio; generic subset guarantees leave its quantifier order unresolved.